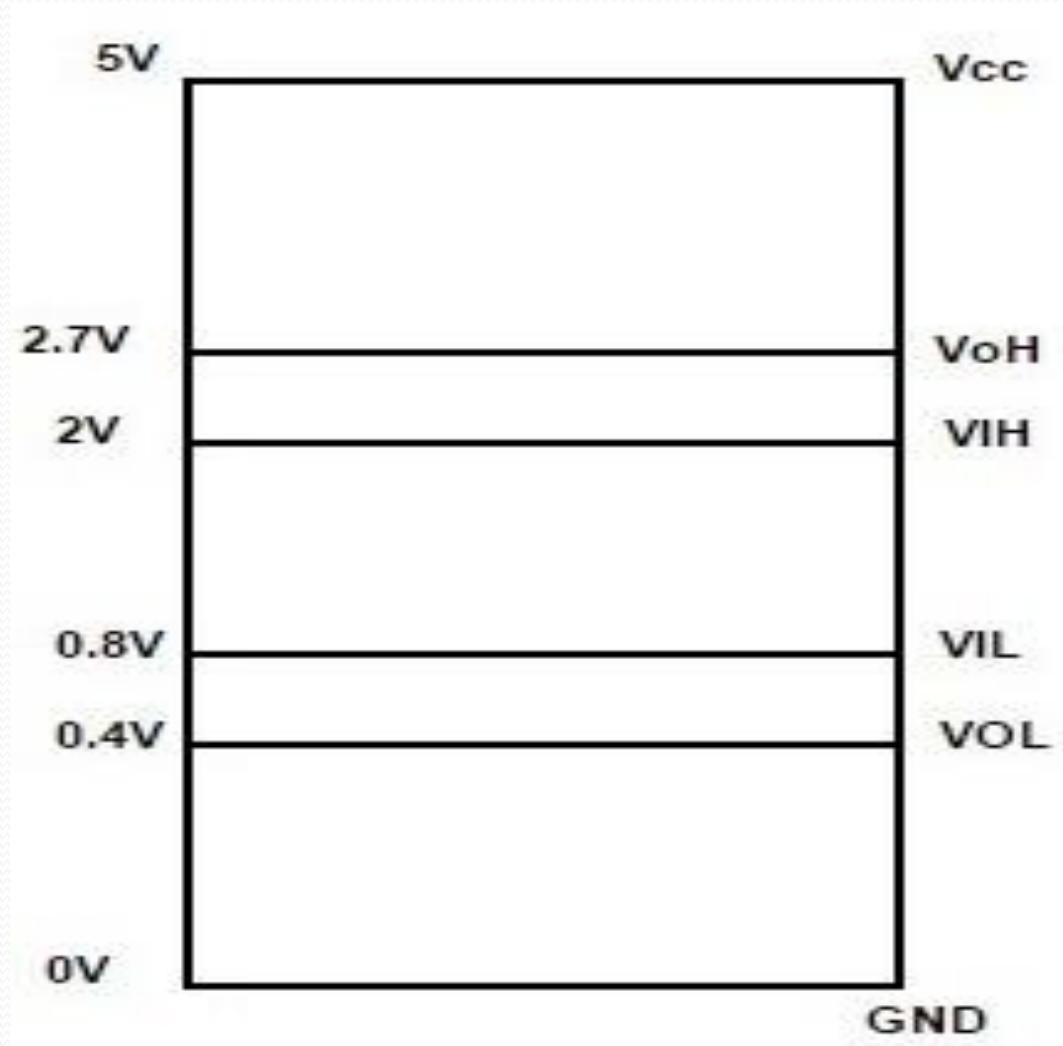


Digital Logic Design

Fundamental concepts

- Modern digital computers, calculators, communication systems, internet, email etc are operating based on the principles of digital techniques.
- These systems are *digital systems*.
- Digital systems work with *Digital signals*
- Digital signals have two discrete values or levels. Low and High
- Digital circuits are used to process these digital signals

Digital signal level representation



In positive logic representation Bit 1 represents Logic high and Bit 0 represent a Logic low as shown in fig 2 a and b. High is represented by +5 Volts and low is represented by -5 Volts or 0 Volts.

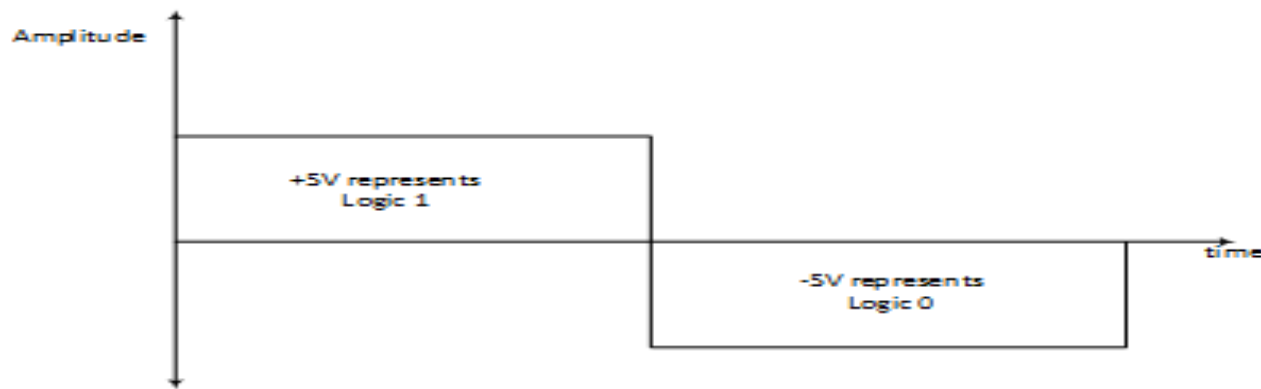


Fig. 2 a. Positive logic representation

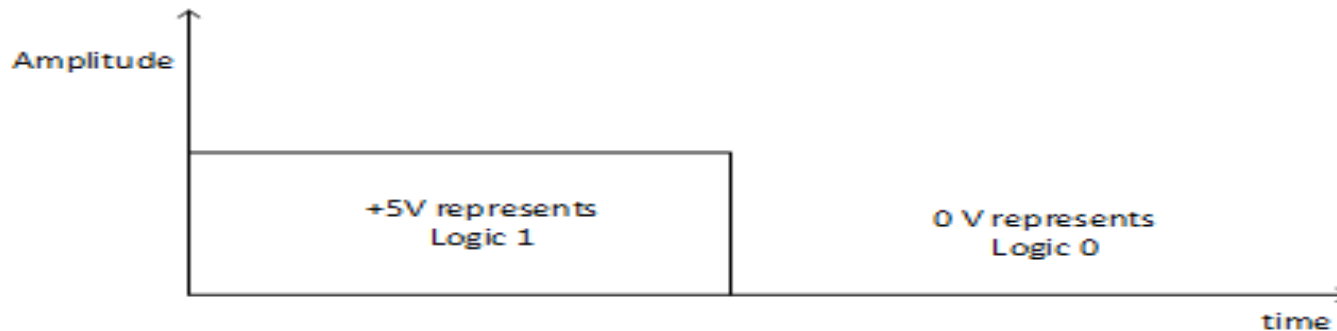


Fig. 2 b. Positive logic representation

In Negative logic representation Bit 1 represents logic low and Bit 0 represents logic high as shown in Fig 3 a and b. In terms of voltage level, bit 1 can be represented as +5V and bit 0 can be represented as 0 V or -5 Volts

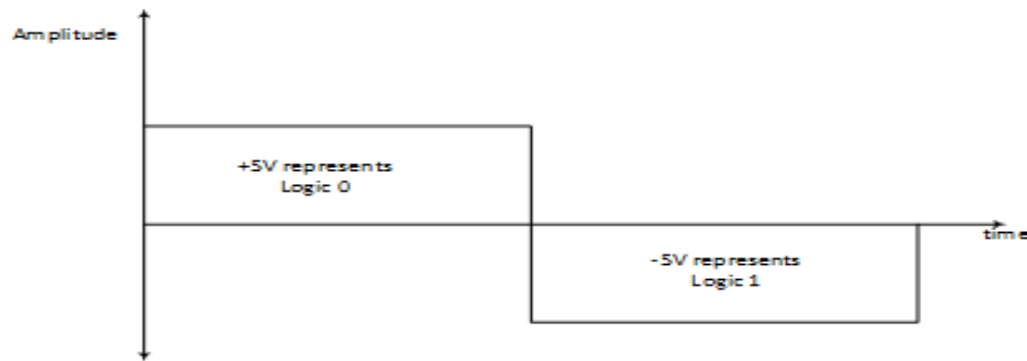


Fig. 3 a. Negative logic representation

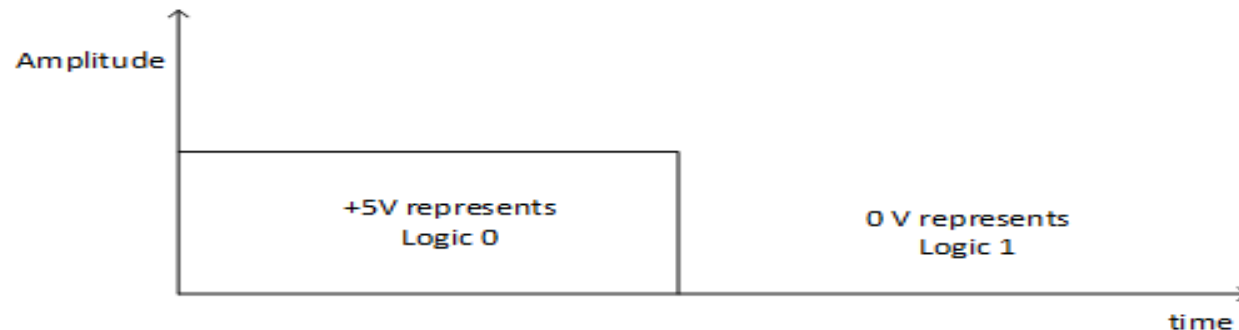


Fig. 3 b. Negative logic representation

- Two discrete signal levels can also be represented by binary digits 1 and 0.
- Binary digit is referred as a *bit*
- Binary number system is used for analysis and design of digital systems
- Two levels can be also designated as ON and OFF or TRUE and FALSE
- George Boole developed binary number systems and Boolean algebra for logic circuits design and analysis

Advantages of digital signal system

- Inexpensive ICs can be used with few external components.
- Operate in one of the two **states, known as ON and OFF makes it very simple.**
- Only a few basic operations are required and are very easy to understand.
- Digital techniques deal with simple logic mathematics called Boolean algebra.

Advantages of digital signal system cond...

- Operation and network analysis of digital circuitry require simple basic concepts like switching speed and loading on the other hand, analysis of analog circuitry (Involved frequency and time domain) are quite complicated.
- Information can be stored for short periods or indefinitely.
- Data can be used for precise calculations.
- Systems can be designed more easily using compatible digital logic families.

Advantages of digital signal system cond...

- Systems can be programmed which show some manner of 'intelligence'. A number of programmable ICs are also available.
- The display of data and other information is very convenient, accurate and elegant by using digital techniques.
- Digital circuits have capability of memory, which makes these circuits highly suitable for computers, calculators, watches, telephones, medical diagnostics, etc.
- To learn programming of digital computers it is worth knowing, the way the digital hardware works.

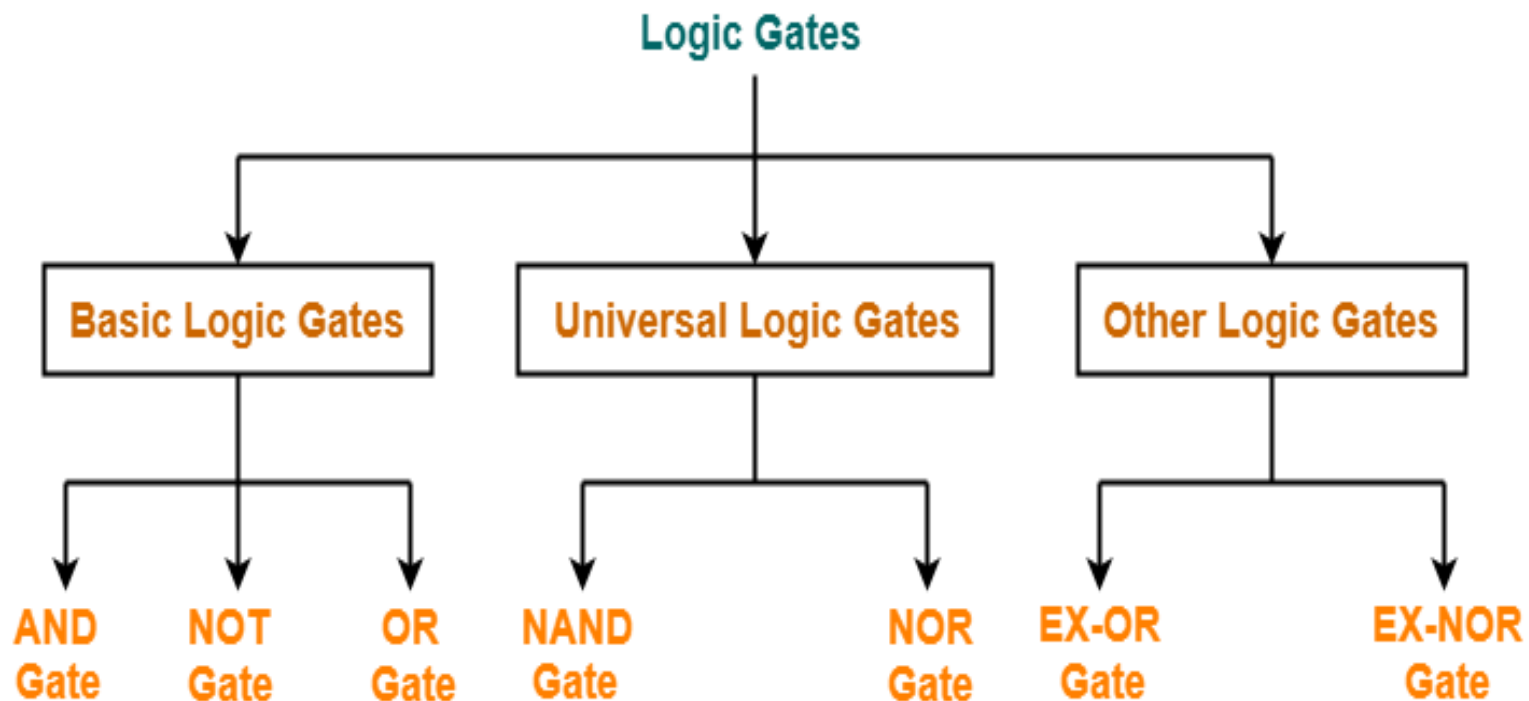
Disdvantages of digital signal system

- The limitations of digital circuitry are as follows:
- Most 'real-world' events are analog in nature.
- Analog processing is usually simpler and faster.
- Digital circuits are appearing in more and more products primarily because of
- low-cost, reliable digital ICs. Other reasons for their growing popularity are accuracy,
- added stability, computer compatibility, memory

Basic Digital Gates

- Basic operations performed in digital systems are AND, OR, NOT and FLIP-FLOP
- Digital circuits which perform these logical operations are known as Logic Gates
- Logic gates have one or more than one number of inputs but only one output.
- Basic Logic gates are : AND, OR, NOT, EX-OR, EX-NOR

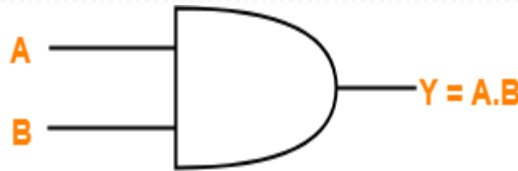
Classification of logic gates



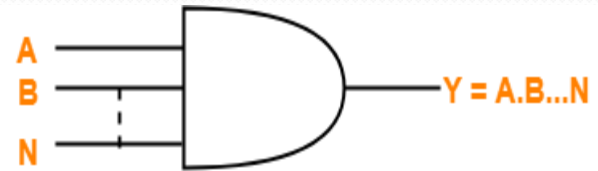
Types of Logic Gates

AND Gate

- Symbol:



2-Input AND Gate



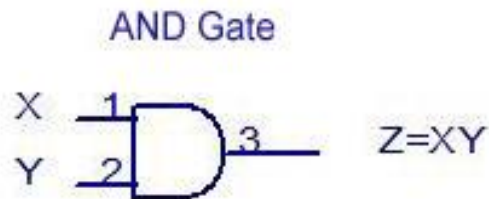
N-Input AND Gate

- It has N inputs ($N \geq 2$) and one output
- Digital signals are applied at the input terminals A, B, ..N
- Output of AND gate is 1 if and only if all the inputs are 1
- Mathematically it is written as
$$Y = A \text{ AND } B \text{ AND } C \dots \text{AND } N$$
$$= A.B.C \dots N$$
$$= ABC \dots N \text{----- } 1$$
- Equation 1 is known as Boolean equation or logical equation of AND gate

Truth Table of AND gate

- Truth table contains all possible input combinations (2^N combinations for N inputs) and their corresponding outputs

2 Input AND Gate

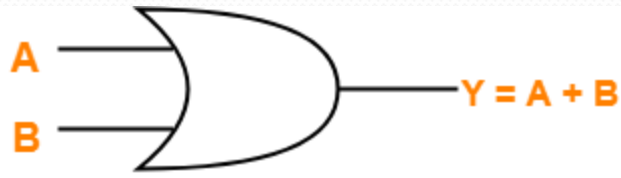


TRUTH TABLE

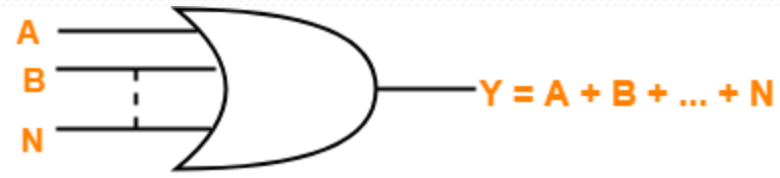
INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate

- Symbol :



2-Input OR Gate



N-Input OR Gate

- Output of an OR gate is 1 if and only if one or more inputs are 1
- Logical equation is given by
$$Y = A \text{ OR } B \text{ OR } C \dots \text{OR } N$$
$$= A + B + C \dots + N$$

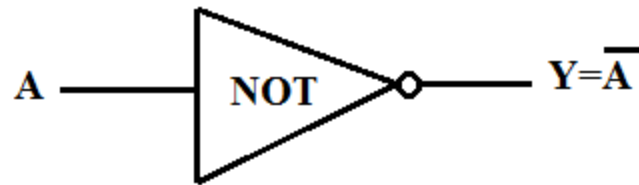
Truth table



Inputs		Output
A	B	$Y=A+B$
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate

- Symbol and Truth table



Input	Output
A	$Y = \bar{A}$
0	1
1	0

- It is also known as an Inverter.
- $Y = \text{NOT } A$
 $= \bar{A}$
- Presence of a small circle, known as bubble is used to indicate inversion in digital circuits

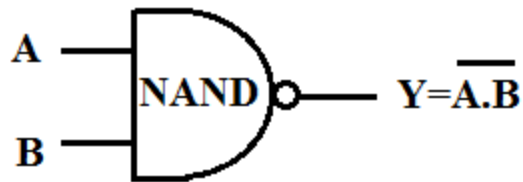
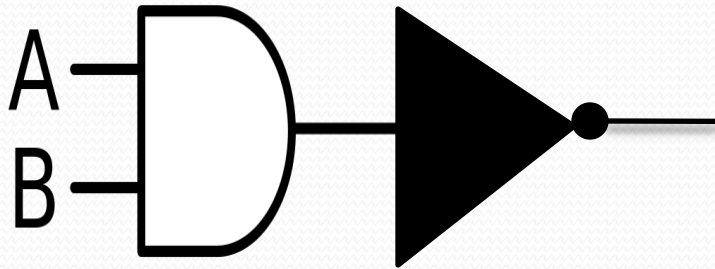
NAND gate

- The NAND gate is a gate which has two or more input terminals and one output terminals.
- The NAND gate is a cascade combination of AND – NOT gates.
- The output of NAND gate is high or 1 when any one or both the inputs are low or 0 and the output is low or 0 when both inputs are high or 1
- Mathematical equation is written as

$$Y' = AB.....N$$

$$Y = \overline{Y'} = \overline{AB.....N}$$

Symbol and truth table of NAND gate



Inputs		Output
A	B	$Y = \overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

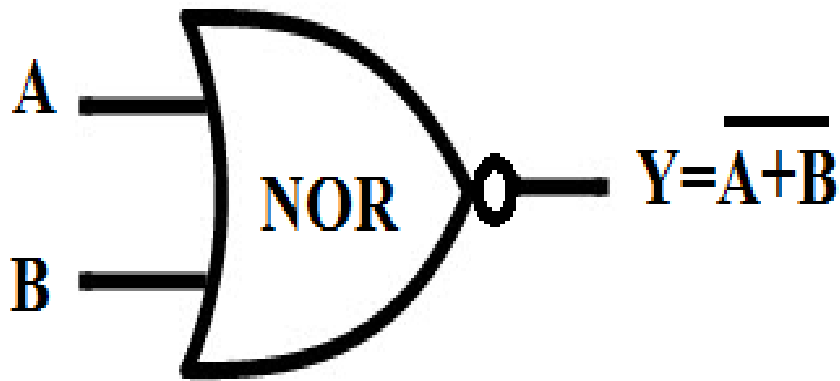
NOR gate

- The NOR gate is a gate which has two or more input terminals and one output terminals.
- The NOR gate is a cascade combination of OR – NOT gates.
- The output of NOR gate is high or 1 when both the inputs are low or 0 and the output is low or 0 when one or both inputs are high or 1.
- Mathematical equation is written as

$$Y' = A + B + \dots + N$$

$$Y = \overline{Y'} = \overline{A + B + \dots + N}$$

Symbol and truth table of NOR gate

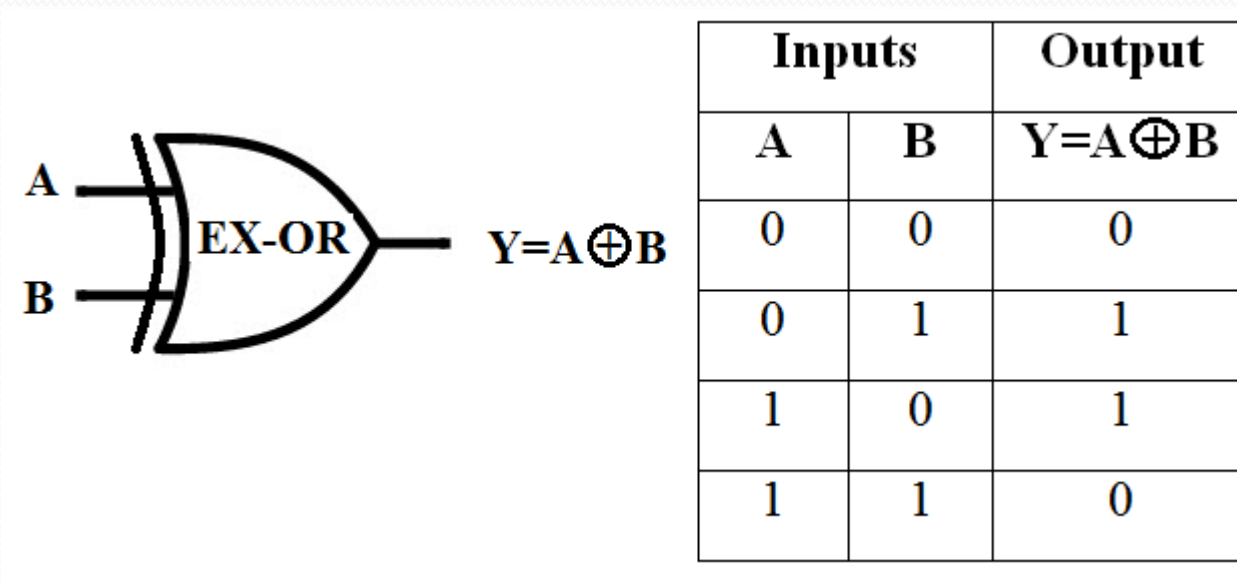


Inputs		Output
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

Ex-OR gate

- The Exclusive gate is a gate which is used in digital data processing circuits.
- This gate has two or more input terminals and one output terminal.
- The EX-OR Gate has the output only high when an odd number of inputs are high and the output is low when both the inputs are low and both the inputs are high.

The symbol of EX-OR Gate & Truth Table



Exclusive- NOR (EX-NOR) GATE

- The EX-NOR gate has terminals like OR gate i.e, two or more input terminals and one output terminal.
- The EX-NOR Gate has the output only high when both the inputs have the same values either high or low and the output is low when an odd number inputs are high.



Inputs		Output
A	B	$Y = A \oplus B$
0	0	1
0	1	0
1	0	0
1	1	1

Alternative logic gates

Original Gates	Alternate Gates
	
	
	
	
	
	

Universal Gates

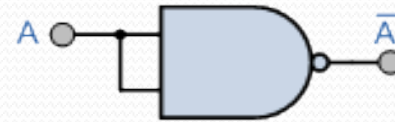
- Any Boolean expression can be realized by using the AND, OR and NOT gates discussed previously.
- From these three operations, two more operations have been derived: the NAND operation and NOR operation.
- These operations have become very popular and are widely used, the reason being the only one type of gates, either NAND or NOR are sufficient for the realization of any logical expression. Because of this reason NAND and NOR gates are known as universal gates.

NAND gate as universal gate

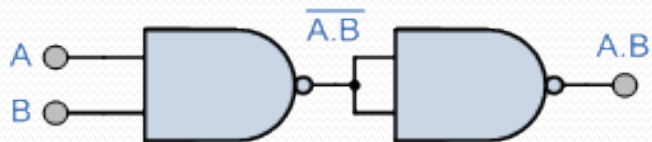
NAND Gate Symbol



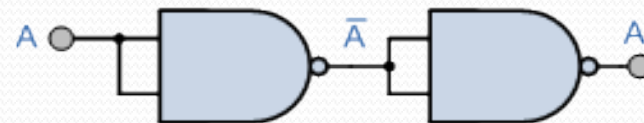
NOT Gate (Inverter)



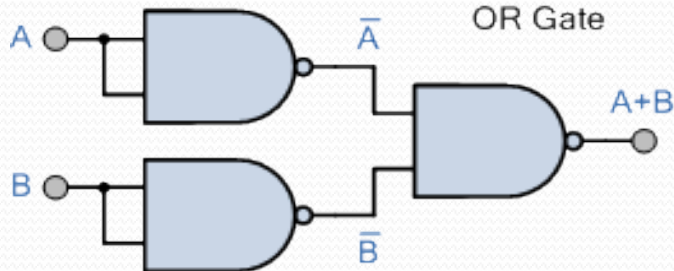
AND Gate



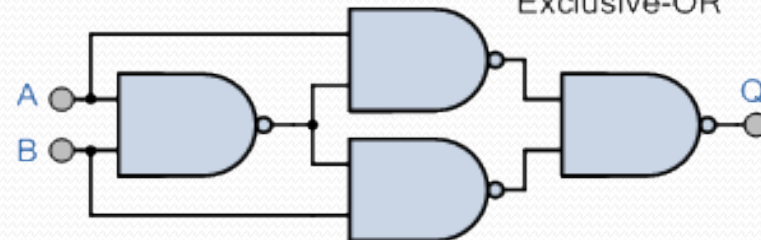
Buffer



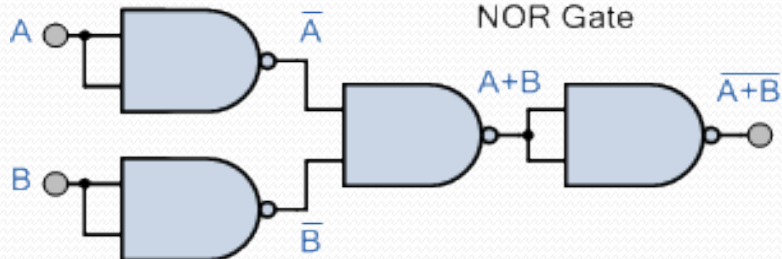
OR Gate



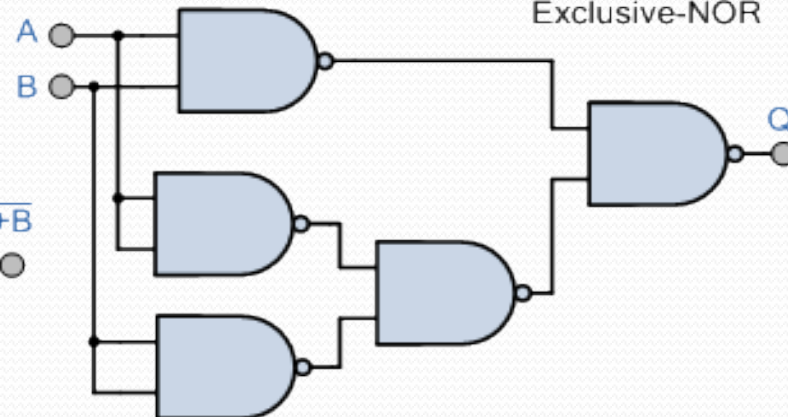
Exclusive-OR



NOR Gate

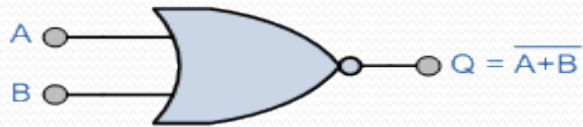


Exclusive-NOR

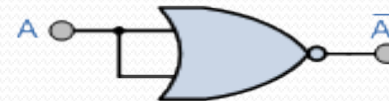


NOR gate as universal gate

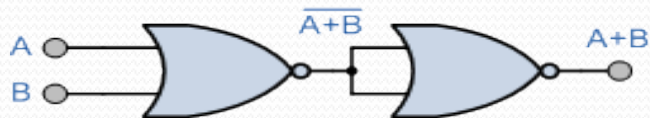
NOR Gate Symbol



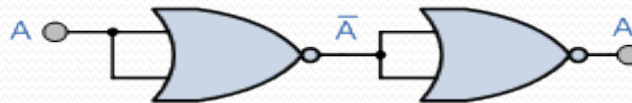
NOT Gate (Inverter)



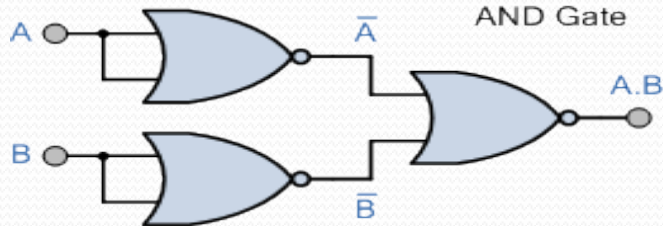
OR Gate



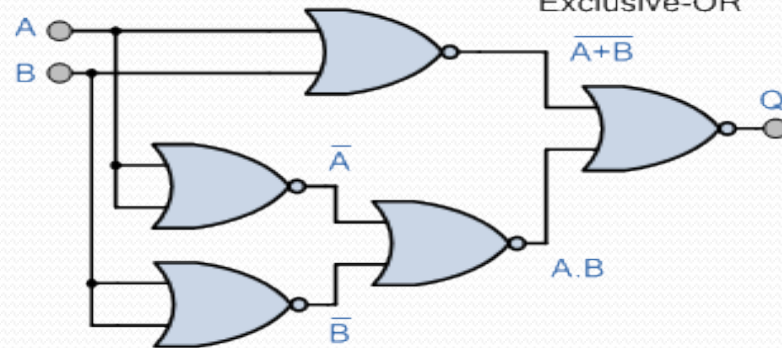
Buffer



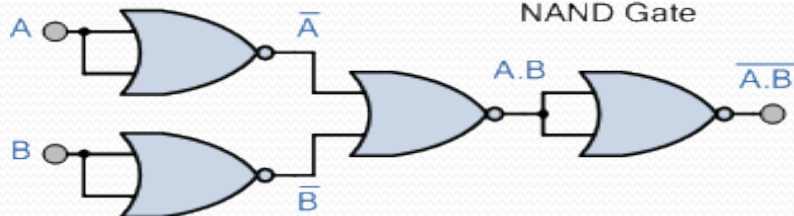
AND Gate



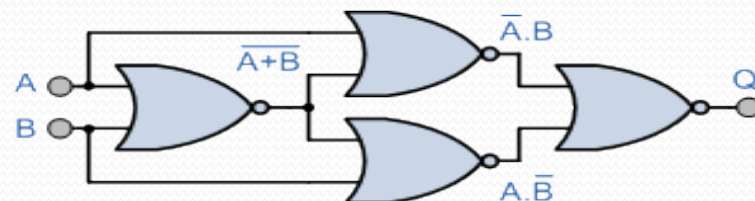
Exclusive-OR



NAND Gate



Exclusive-NOR



Boolean Algebra Laws

Basic Boolean algebra laws

	AND	OR
Commutative	$A \cdot B = B \cdot A$	$A + B = B + A$
Associative	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$	$(A + B) + C = A + (B + C)$
Distributive	$A \cdot (B + C) = A \cdot B + A \cdot C$	$A + (B \cdot C) = (A + B)(A + C)$
De Morgan's Theorem	$(A \cdot B)' = A' + B'$	$(A + B)' = A' \cdot B'$
	$(A \cdot B \cdot C)' = A' + B' + C'$	$(A + B + C)' = A' \cdot B' \cdot C'$
	$(A \cdot B \cdot C \cdot \dots)' = A' + B' + C' + \dots$	$(A + B + C + \dots)' = A' \cdot B' \cdot C' \cdot \dots$
	$A \cdot 0 = 0$	$A + 0 = A$
	$A \cdot 1 = A$	$A + 1 = 1$
	$A \cdot A = A$	$A + A = A$
	$A \cdot A' = 0$	$A + A' = 1$
	$A \cdot (A + B) = A$	$A + (A \cdot B) = A$
	$A \cdot (A' + B) = A \cdot B$	$A + (A' \cdot B) = A + B$
	$A \cdot B + A \cdot B' = A$	$(A + B)(A + B') = A$
	$(A')' = A$	$0' = 1$

Images of logic gate ICs



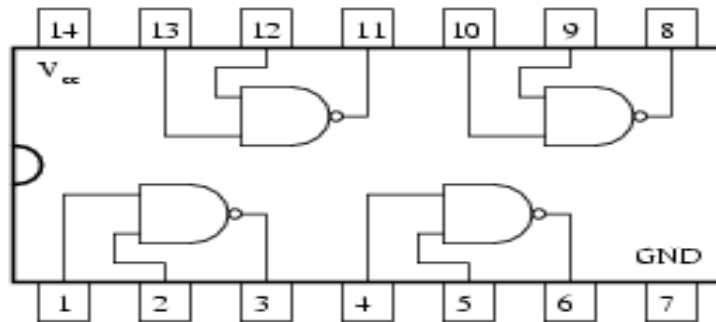
Logic gates

- If you are interested in digital electronics you 'd like to use logic gates.
- Mentioned in this table 74xx TTL logic family part numbers.

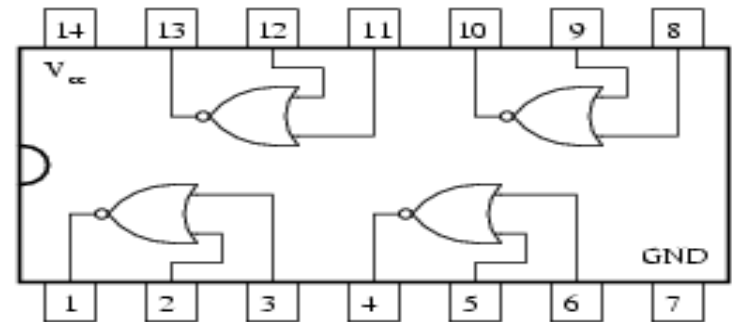


Part number	Description
7400	NAND
7402	NOR
7404	NOT
7408	AND
7432	OR
7486	XOR
7474	D-flip flop
7476	J-K flip flop
7448,7447	7 segment driver
7407	Open collector buffer

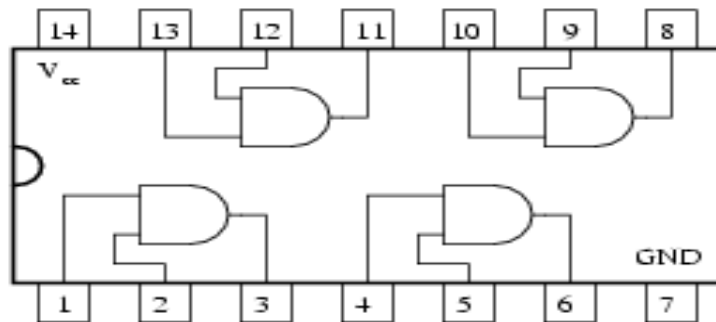
5400/7400
Quad NAND gate



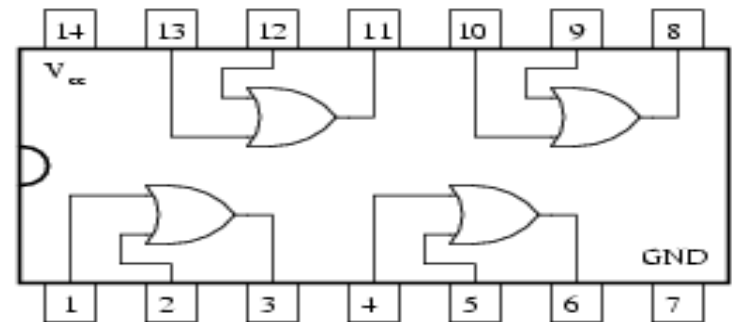
5402/7402
Quad NOR gate



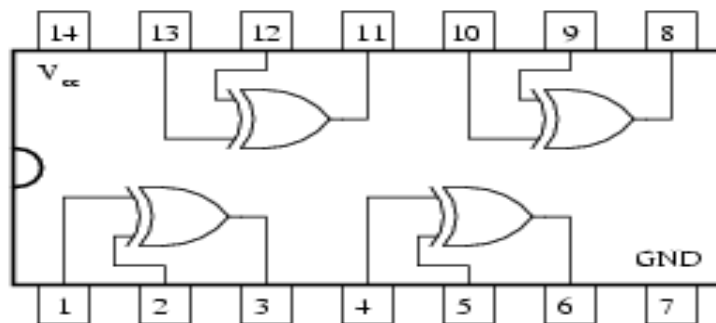
5408/7408
Quad AND gate



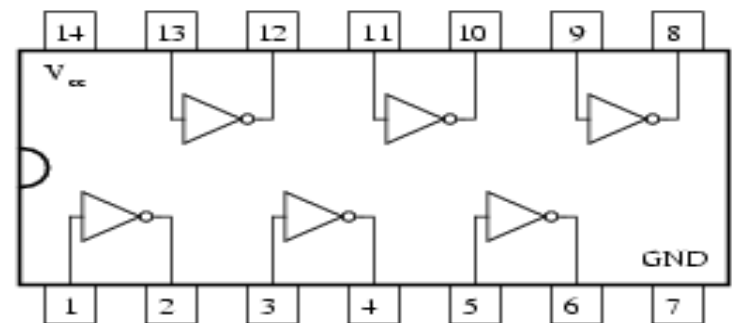
5432/7432
Quad OR gate



5486/7486
Quad XOR gate



5404/7404
Hex inverter





THANK YOU !!!